GLOBAL SECTIONS AND DERIVED HOM AGAINST A PERFECT OBJECT IN THE EXISTENCE SITUATION ARE PERFECT AND COMPUTE THE REDUCTIONS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file flat.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-flat/retained-preamble.tex), a verbatim excerpt of the native Xy-pic setup and of the native theorem declarations and mathematical macros, followed by the project own definitions of the reference, slogan and history display environments (verbatim package plus three newenvironment lines) that the retained file carries at its end and that the project itself typesets as comments. No macro was invented or redefined, and the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 0CTE (sources/stacks-flat/pinned-original-source.tex line 8079).

Allowed standard prerequisites (editorial).

- The object K and the pendant lemma. Same chapter, Lemma labelled lemma-compute-what-it-should-be, invoked in the statement. Supplies K and the system F_n used throughout.
- Base change for derived Hom of perfect complexes. Derived Categories of Schemes, Lemma labelled perfect-lemma-base-change-RHom-perfect, cited at proof lines 8104-8106. Shows each N_n is perfect over D(A_n) with compatible base change.
- Derived limits of perfect complexes. More on Algebra, Lemma labelled more-algebra-lemma-Rlim-perfect-gives-perfect, cited at proof lines 8110-8114. Makes Rlim N_n perfect and recovers its base changes.
- Derived Hom as global sections of the sheaf Hom, and duality with the tensor product formulation. Cohomology, Section labelled cohomology-section-global-RHom, cited at proof lines 8100-8102, and Lemma labelled cohomology-lemma-dual-perfect-complex, cited at proof lines 8123-8125. Identifies the functors and translates statement (1) into statement (2).

Source reference. Original reference 0CTC: sources/stacks-flat/pinned-original-source.tex, source lines 8036–8036. Same-chapter pointer into the author environment situation at source line 8036; invoked at proof line None of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0CTD: sources/stacks-flat/pinned-original-source.tex, source lines 8062–8062. Same-chapter pointer into the author environment lemma

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at source line 8062; invoked at proof line None of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original section 0B6A: sources/stacks-chow/tags.txt, source lines 14063–14063. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 8103 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0A1J: sources/stacks-chow/tags.txt, source lines 12672–12672. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 8105 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0CQG: sources/stacks-chow/tags.txt, source lines 15961–15961. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 8110 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 08DQ: sources/stacks-chow/tags.txt, source lines 10645–10645. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 8124 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1. *In Situation 0CTC let K be as in Lemma 0CTD. For any perfect object E of $D(\mathcal{O}_X)$ we have*

1. *$M = R\Gamma(X, K \otimes^{\mathbf{L}} E)$ is a perfect object of $D(A)$ and there is a canonical isomorphism $R\Gamma(X_n, \mathcal{F}_n \otimes^{\mathbf{L}} E|_{X_n}) = M \otimes^{\mathbf{L}}_A A_n$ in $D(A_n)$,*
2. *$N = R\mathrm{Hom}_X(E, K)$ is a perfect object of $D(A)$ and there is a canonical isomorphism $R\mathrm{Hom}_{X_n}(E|_{X_n}, \mathcal{F}_n) = N \otimes^{\mathbf{L}}_A A_n$ in $D(A_n)$.*

In both statements $E|_{X_n}$ denotes the derived pullback of E to X_n.

Proof. Proof of (2). Write $E_n = E|_{X_n}$ and $N_n = R\mathrm{Hom}_{X_n}(E_n, \mathcal{F}_n)$. Recall that $R\mathrm{Hom}_{X_n}(-,-)$ is equal to $R\Gamma(X_n, R\mathcal{H}om(-,-))$, see Cohomology, Section 0B6A. Hence by Derived Categories of Schemes, Lemma 0A1J we see that N_n is a perfect object of $D(A_n)$ whose formation commutes with base change. Thus the maps $N_n \otimes^{\mathbf{L}}_{A_n} A_{n-1} \to N_{n-1}$ coming from φ_n are isomorphisms. By More on Algebra, Lemma 0CQG we find that $R\lim N_n$ is perfect and that its base change back to A_n recovers N_n. On the other hand, the exact functor $R\mathrm{Hom}_X(E, -) : D_{QCoh}(\mathcal{O}_X) \to D(A)$ of triangulated categories commutes with products and hence with derived limits, whence

$$R\mathrm{Hom}_X(E, K) = R\lim R\mathrm{Hom}_X(E, \mathcal{F}_n) = R\lim R\mathrm{Hom}_X(E_n, \mathcal{F}_n) = R\lim N_n$$

This proves (2). To see that (1) holds, translate it into (2) using Cohomology, Lemma 08DQ. □